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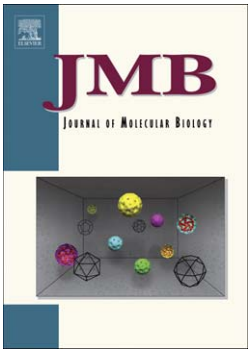

The HADDOCK2.2 web server: User-friendly integrative modeling of biomolecular complexes

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The HADDOCK2.2 web server: User-friendly integrative modeling of biomolecular complexes

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Abstract

The prediction of the quaternary structure of biomolecular macromolecules is of paramount importance for fundamental understanding of cellular processes and drug design. In the era of integrative structural biology, one way of increasing the accuracy of modelling methods used to predict the structure of biomolecular complexes is to include as much experimental or predictive information as possible in the process. This has been at the core of our information-driven docking approach HADDOCK. We present here the updated version 2.2 of the HADDOCK portal, which offers new features such as support for mixed molecule types, additional experimental restraints and improved protocols, all of this in a user-friendly interface. With well over 6000 registered users and 108000 jobs served, an increasing fraction of which on grid resources, we hope this timely upgrade will help the community to solve important biological questions and further advance the field. The HADDOCK2.2 web server is freely accessible to non-profit users at http://haddock.science.uu.nl/services/HADDOCK2.2.

Keywords

Biomolecular docking; Data-driven; Hybrid modeling; Protein complexes; Grid computing.

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Introduction

Cellular metabolism is a highly regulated and adaptive system where proteins, the main participants, form a vast network of interactions collectively known as the interactome. Knowledge of the three dimensional (3D) atomic structure of protein-protein interactions is therefore critical for a fundamental understanding of cellular and molecular biology, as well as for rational drug-design. Unfortunately, solving such structures using classical high-resolution methods (X-ray crystallography and NMR spectroscopy) is not trivial, as each has its own limitations (e.g. protein flexibility, size, strength of the interaction). Considering the magnitude of the interactome, complementary high-throughput methods such as computational docking are necessary if we aim to close the structure gap [1]. The goal of protein-protein docking is to predict the structure of a complex starting from the individual structures of its components [2], which can either be experimentally determined or predicted [3].

Despite continuous advances in the field, the accuracy of *ab initio* docking – without using any experimental restraints – remains generally low [4]. Data-driven approaches such as HADDOCK [5,6], which integrate information derived from biochemical, biophysical or bioinformatics methods to enhance sampling, scoring, or both [2], perform remarkably better. The information that can be integrated is quite diverse: interface restraints from NMR, mutagenesis experiments, or bioinformatics predictions [7,8]; shape data from small-angle X-ray scattering [9] and cryo-electron microscopy experiments [10]; and orientations of the individual structures in the complex from NMR residual dipolar couplings [11], relaxation anisotropy [12] and pseudocontact shifts experiments [13]. The potential of data-driven docking is reflected in the success of the HADDOCK server and software in recent CAPRI experiments (Critical Assessment of Protein Interaction) [14,15], as well as in the number of structures deposited (>120) in the Protein Data Bank (PDB), which were calculated using our software.

Five years ago, we introduced the HADDOCK web server to provide a user-friendly interface to the software and streamline its usage by non-expert users in the structural biology field [16]. Shortly after, it was updated to handle multi-body docking [17]. The development of new and improved protocols and the inclusion of additional sources of restraints culminated in the recently released version 2.2 of the software, followed by an update of the web server interfaces. Throughout the next section, we will provide an overview of the newly updated HADDOCK web server and discuss the most relevant additions. The server is freely accessible at

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http://haddock.science.uu.nl/services/HADDOCK2.2 to non-profit users upon registration. We conclude by presenting usage statistics of the server to demonstrate the usefulness and power of providing easy and free access to scientific software.

Overview and advances

The HADDOCK web server was created to facilitate the use of our docking software, by removing the burden of its installation and setup, as well as by providing validation routines for input data and options. In addition, since HADDOCK runs are computationally demanding, the web server offers the users access to sufficient resources - our local cluster(s) - to complete their runs within a few hours. A grid-enabled version of the server can be accessed via the WeNMR web site (www.wenmr.eu) [18], which uses resources provided by the European Grid Initiative (EGI) and the associated National Grid Initiatives (NGIs). This setup, which currently handles most submissions, provides more than 110.000 CPU cores distributed over 41 sites worldwide (see http://gstat.egi.eu/gstat/geo/openlayers#/VO/enmr.eu).

The HADDOCK web server aggregates seven different interfaces, each associated with a different level of control over the docking protocol reflected by the number of parameters that can be changed. New users are granted access to the *Easy* and the associated *Prediction Interface* only, but can request access to the *Expert* and *Guru* levels, and their associated interfaces, if necessary.

The *Easy* interface provides the most basic level of control. It allows the user to either upload two structures in PDB format or download them directly from the RCSB PDB, and define sets of active and passive residues that represent the (putative) interface. Unlike previous versions, HADDOCK 2.2 supports single (protein, small molecule, RNA, or DNA) and mixed (protein-DNA, protein-RNA) molecule types. This was implemented to handle the docking of proteins onto a nucleosome complex - a recent CAPRI target.

The *Expert* interface builds on the *Easy* interface and allows the user to manually specify the protonation state of each histidine residue in the proteins, which is otherwise determined automatically with MolProbity [19]. Also, it offers control over which regions of the molecules are semi-flexible and fully flexible segments, which has an impact during the refinement stage of the docking. Lastly, the user is given the option to define the charge state of the N- and C-terminus of the protein. The *Expert* interface also provides a *Distance Restraints* section, where the user has the option to upload user-defined ambiguous and unambiguous restraints files and/or use center-of-mass restraints, useful for blind or *ab initio* docking when no other information is available, but also to ensure compactness of the generated models. The center-of-mass restraints are automatically generated

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by calculating the dimensions of each molecule along the x, y and z-axis (d_x, d_y, d_z) and summing the average of the two smallest components per molecule. The resulting distance is used to define a restraint between the center of mass of each subunit with an additional upper bound correction of 1Å [9]. In addition, the *Expert* interface gives control over the *Sampling Parameters*, including namely the number of structures to generate at each stage and whether or not to perform solvated docking [20,21,23]. Finally, it exposes the *Clustering Parameters* that define the clustering algorithm and cutoff. In version 2.2, in addition to RMSD-based clustering, there is the option of using the Fraction of Common Contacts (FCC) clustering algorithm [23], which is significantly faster and especially useful for symmetric complexes.

The *Guru* interface gives full access and control to ~500 parameters, nearly all that are available in HADDOCK. The *Distance Restraints* section now offers a new radius of gyration restraint, information that can be extracted, for example, from SAXS experiments. *Non-crystallographic Symmetry Restraints* and *Symmetry Restraints* are also available at this level and have been extended to handle C4- and D2-symmetries in addition to the already available C2-, C3- and C5-symmetries. There are additional sections for other types of NMR-based restraints, such as *Residual Dipolar Couplings* [11], *Relaxation Anisotropy* [12], and the recently added *Pseudo Contact Shifts* [13]. These latter require a tensor distance restraints file and the definition of the rhombic and axial components of the anisotropic tensor. Besides the restraints, all the energy evaluations, scoring functions and analysis parameters can be tweaked; advanced parameters for the sampling protocols are also available at this level, offering a greater degree of control, for example, on the extent of each refinement stage. There are also dedicated options to the solvated docking protocol, which now uses by default propensities based on the Kyte-Doolittle hydrophobicity scale, as these have been shown to improve the protocol [21]. The original statistical-based propensities [20], recently expanded to include nucleotides [22], can still be selected via a dropdown menu.

The remaining four interfaces consist of: the *Prediction Interface*, which is similar to the Easy interface, but with settings geared towards using bioinformatics interface predictors such as CPORT [7]; the *Refinement Interface* (expert-level access), which runs only the water refinement stage on the uploaded structures and can be used for scoring purposes; the *Multi-body Interface*, based on the *Guru* interface, supporting upload of up to six molecules that will be docked simultaneously [17] and also featuring the *Molecule Interaction Matrix* section. This new addition displays a table with scaling factors to adjust the interaction forces between different subunits, allowing molecules to become invisible to each other during the docking, which is useful in cases where multiple binding modes are required to satisfy the experiment data (see for example [24]). The use of

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ambiguous interaction restraints within this interface requires the user to upload a restraints table file in the *Distance Restraints* section This requirement of uploading distance restraint files instead of supplying residue lists as in the other interfaces was meant to make users think carefully about their system since in multi-body docking multiple interfaces will be defined that might not all be supposed to interact. To facilitate the creation of custom ambiguous interaction restraint files between any number of molecules, an interface called *Gentbl* was created. Finally, the web server also offers a *File Upload Interface* to allow the user to upload a run parameter file, created upon successful validation and submission to the queue, and thus easily redo a docking run or re-run it with slight changes in the parameters.

At submission time, once the input data have been properly validated, the server offers the option to download a parameter file, and provides a link to the results page, which is also emailed to the user. Users are encouraged to save the parameter file since it contains all required input data and settings to reproduce the docking, as recommended in the *"Outcome of the First wwPDB Hybrid/Integrative Methods Task Force Workshop"* (Recommendation 1) [25]. The results page allows monitoring of the progress of the docking run. After a successful docking run, the user will receive another e-mail redirecting him/her to the updated results page (see Figure 1 for an excerpt of presented results). The page indicates how many structures of the water-refined models could be clustered, and lists the clusters in the order of their HADDOCK score. For each cluster, detailed statistics are displayed, representing the average values calculated over the top four best scoring structures within each cluster. Besides the HADDOCK score and other standard energies (van der Waals, etc.), a z-score has been added. The z-score represents how many standard deviations the HADDOCK score of a given cluster is separated from the mean of all clusters, i.e. the lower the z-score, the better. To visualize the results, plots are displayed at the bottom of the results-page, showing for example the HADDOCK score of all solutions against the interface-ligand-RMSD (i-l-RMSD) compared to the best scoring structure, together with cluster averages and their spreads.

Usage statistics

Since its opening in June 2008, the HADDOCK web server has seen a sustained increase in the number of registrations to reach over 6000 registered users to date distributed all over the world (Figure 2). More than 108000 runs have been processed, 28% of which have run on EGI grid resources. This percentage has increased to 75% for the HADDOCK2.2 server submission. An overview of the number of runs processed per month with

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their distribution over local and grid resources is shown in Figure 2. Since the launch of the HADDOCK2.2 web server in March 2015, an increased fraction of runs are handled by the new 2.2 portal. Statistics over the last year (since January 2014) indicate that the portal is processing on average 75 docking runs per day. The execution wall time averaged over both local cluster and grid resources is around 16 hours per HADDOCK run, as approximately 75 runs are handled per day with up to 50 jobs running in parallel (and a maximum of 5 concurrent jobs per user). The exact run time depends on the size of the system being docked and the parameter settings and can vary between half an hour and several days. The server home page reports both the number of running and pending jobs, allowing users to get an estimate of the waiting time. A majority of the docking runs are dealing with protein-protein and protein-peptide docking (~61%), ~19% correspond to protein-nucleic acids systems and quite a significant fraction (~20%) is dealing with protein-small molecule docking (both small ligand and oligosaccharides). These numbers demonstrate the popularity and widespread usage (both in terms of geographic distribution and type of systems being studied) of our HADDOCK web server.

Acknowledgements

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FIGURE LEGENDS

Figure 1: Excerpts from an example result page of a HADDOCK2.2 docking run.

Figure 2: HADDOCK web server statistics (as of September 1st 2015). The worldwide distribution of registered users (>6000) is shown with a blue-scale color-coding. On the top right, the distribution of the type of systems docked using HADDOCK is shown (taken from a HADDOCK online survey with >325 respondents). The middle bar graph shows the yearly increase in the number of registered users since 2010 and the bottom plot shows the number of processed user runs per month since last year (2014), distinguishing between runs that have been processed on our local resources or on the EGI grid resources, and between the previous and current (2.2) version of the server.

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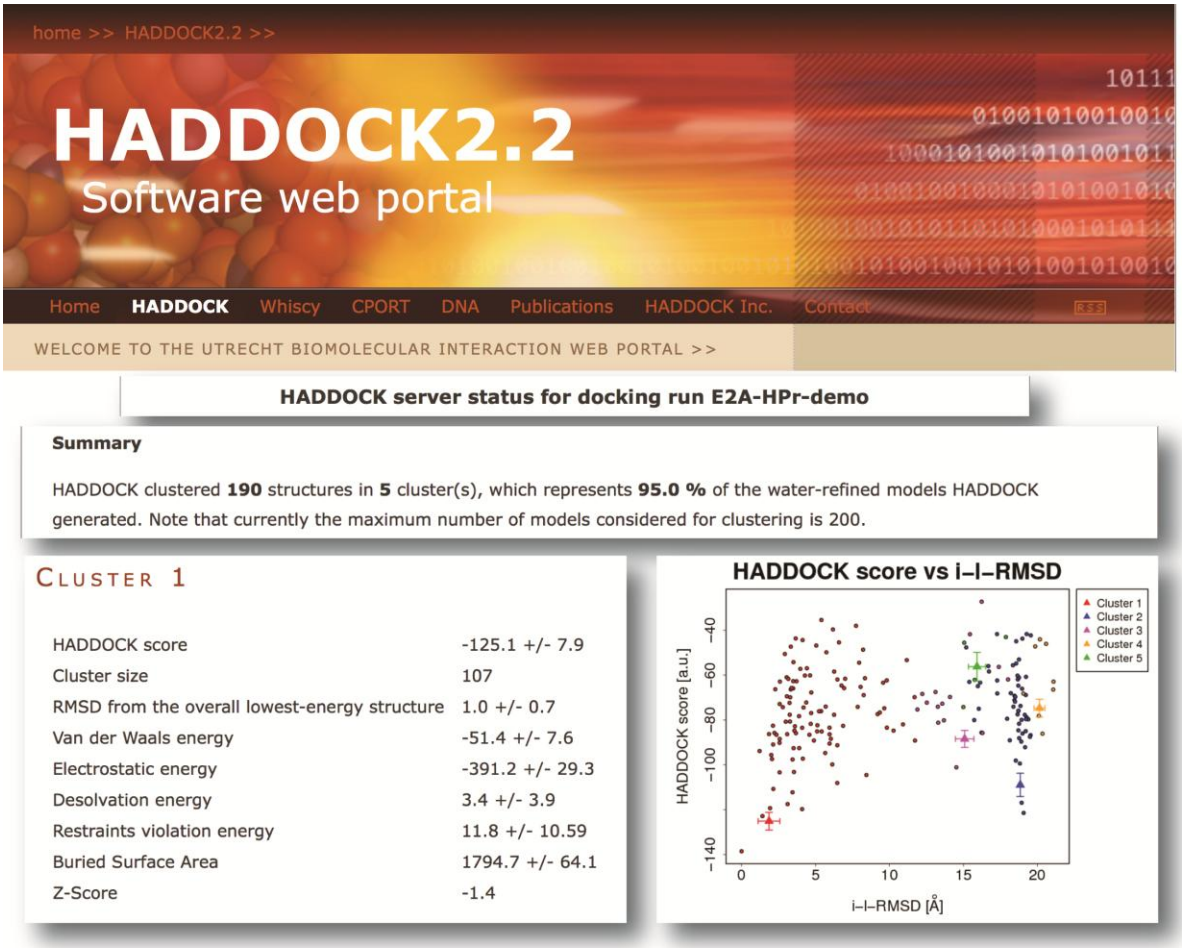

Figure 1

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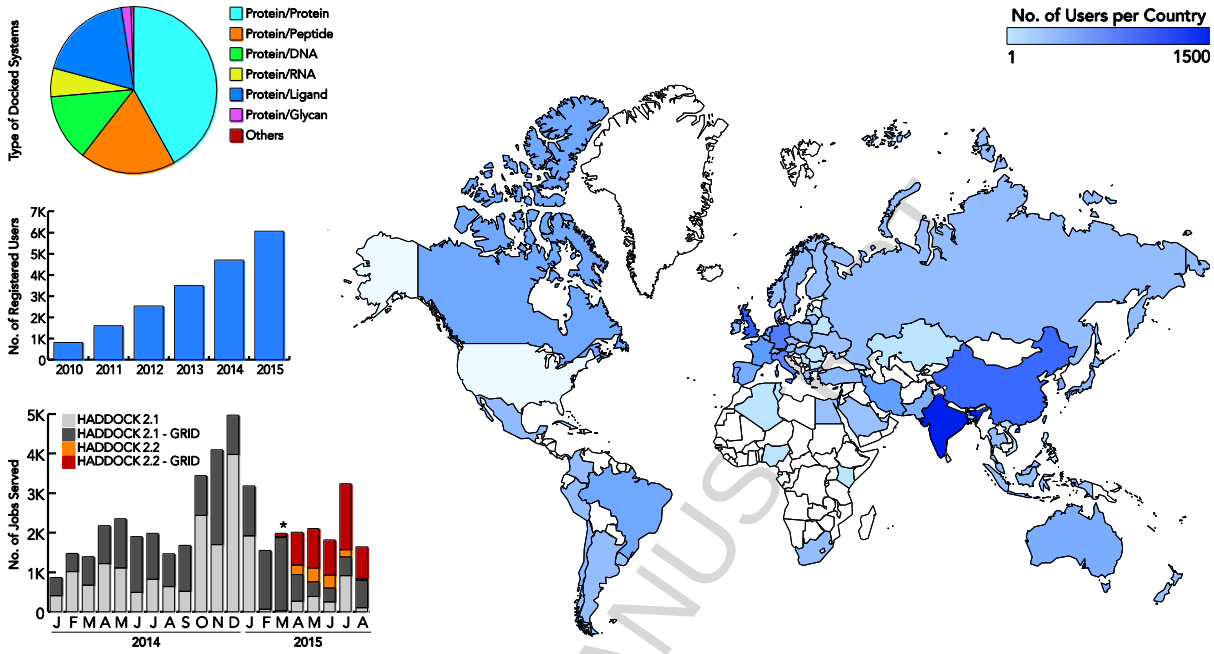

Figure 2

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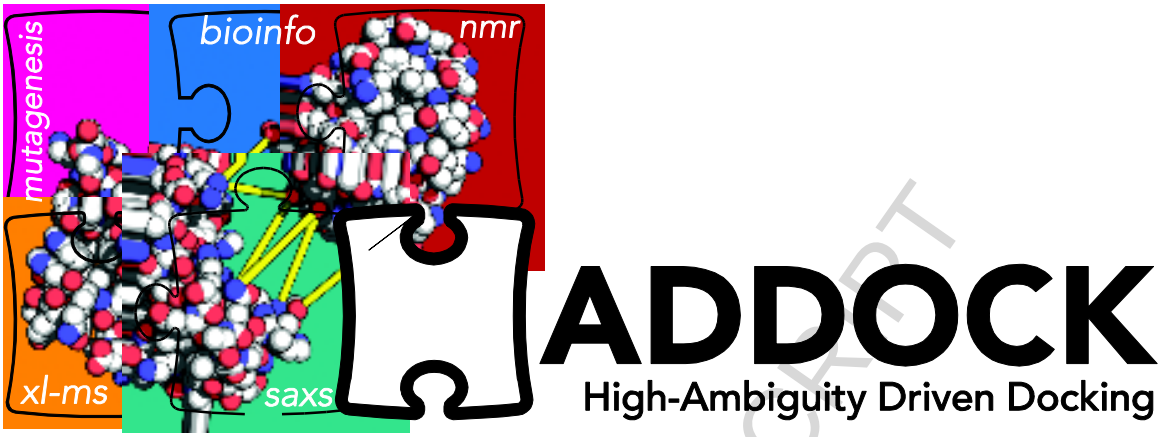

Graphical Abstract

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Highlights

- The HADDOCK webserver has been upgraded to use HADDOCK version 2.2.
- It supports mixed-type molecules, e.g. protein-nucleic acid, such as nucleosomes.
- NMR-based pseudocontact shifts and radius of gyration restraints are included.
- The multibody and solvated docking protocol are extended.
- The HADDOCK webserver counts over 6000 users and has served more than 100000 jobs.

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